EMNLP 2026 Main Conference

Cognition on Graph: Navigating Massive Knowledge Space via Cognitive Cycles and Bidirectional Graph-Text Synergy

Gengxian Zhou[1,2], Jian Xu[3,4], Zichen Tang[1], Shiming Xiang[3,4], Haihong E[1,*], Cheng-Lin Liu[3,4,2,*]

* Corresponding authors.

[1] Beijing University of Posts and Telecommunications [2] Zhongguancun Academy, Beijing

[3] MAIS, Institute of Automation, Chinese Academy of Sciences

[4] School of Artificial Intelligence, University of Chinese Academy of Sciences

EMNLP2026 BUDAPEST

Motivation

Multi-hop QA over global knowledge bases

Answering multi-hop questions requires building evidence chains across large-scale knowledge graphs and text corpora. This involves navigating noisy evidence, ambiguous entities, and missing graph connections.

Gap 1 · Insufficiently adaptive exploration

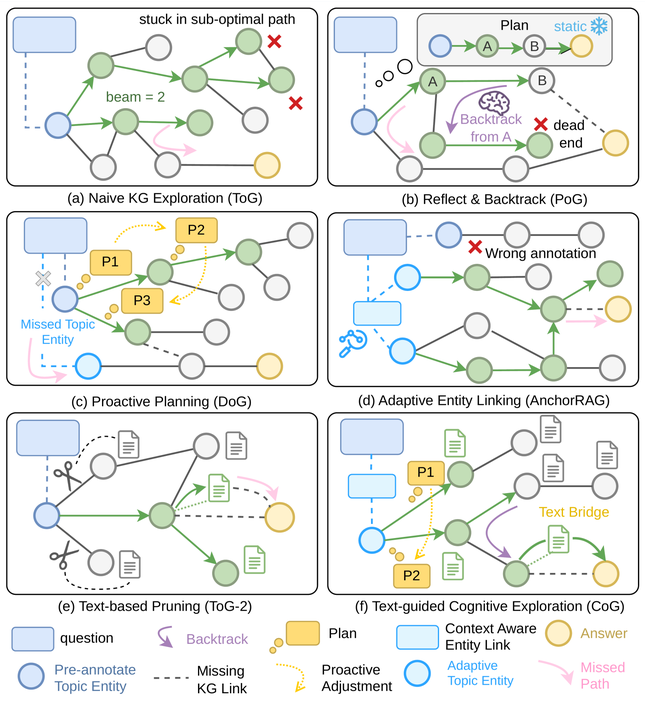
Exploration strategies do not sufficiently adapt to accumulated evidence, making it difficult to recover from irrelevant paths and dead ends.

Gap 2 · Shallow graph–text integration

Text serves as supporting evidence, but entities discovered in text are not fully used to guide graph exploration, limiting the ability to bridge gaps in incomplete KGs.

Idea: Adaptive exploration with graph–text synergy

Adapt the exploration strategy as evidence arrives, and use entities discovered in text as new anchors for graph exploration.



Method	Scope	Text Corpus		Knowledge Graph			Key Capabilities				Exploration Strategy
		#Words	#Ent.	Source	#Ent.	#Trip.	Text Guide	Refl. & Bk.	Pro. Plan.	Adapt. EL.	
VectorRAG	Local	1.5M	21K	–	–	–	–	–	–	–	Dense Retrieval
GraphRAG Series†	Local	0.9M	12K	Constr.‡	97K	1.4M	–	–	–	–	Graph-Guided Retrieval
StructGPT, KG-Agent	Global	–	–	Freebase	39M	1.9B	✗	✗	✗	✗	Interaction History
ToG, ReKnoS	Global	–	–	Wikidata	115M	1.7B	✗	✗	✗	✗	Beam Search
PoG	Global	–	–	Freebase	39M	1.9B	✗	✓	✗	✗	Beam Search w/ Backtrack
DoG	Global	–	–	Freebase	39M	1.9B	✗	✗	✓	✗	Multi-Agent Debate
AnchorRAG	Global	–	–	Freebase	39M	1.9B	✗	✗	✗	✓	Parallel Anchor Exploration
KERAG	Global	–	–	Wikidata	115M	1.7B	✗	✗	✗	✓	Schema-Guided Expansion
ToG-2	Global	5B	7.1M	Wikidata	115M	1.7B	✗	✗	✗	✗	Beam Search
CoG (Ours)	**Global**	**5B**	**7.1M**	**Wikidata**	**115M**	**1.7B**	✓	✓	✓	✓	**Cognitive Cycle**

Table 1: Comparison of retrieval and exploration methods. We report knowledge base scales as total **Words** and **Ent**ity documents for Text, versus **Ent**ities and **Trip**les for KG (Units: K=Thousand, M=Million, B=Billion). **CoG** uniquely operates on a massive scale while integrating all key exploration capabilities: **Text Guide** (Text entity guides graph exploration), **Refl. & Bk.** (Reflection & Backtrack), **Pro. Plan.** (Proactive planning), and **Adapt. EL.** (Adaptive entity linking). †: Includes HippoRAG, HippoRAG2, GFM-RAG, et al. ‡: Graphs constructed from local text corpora. ✓: Present, ✗: Absent, "–": not applicable.

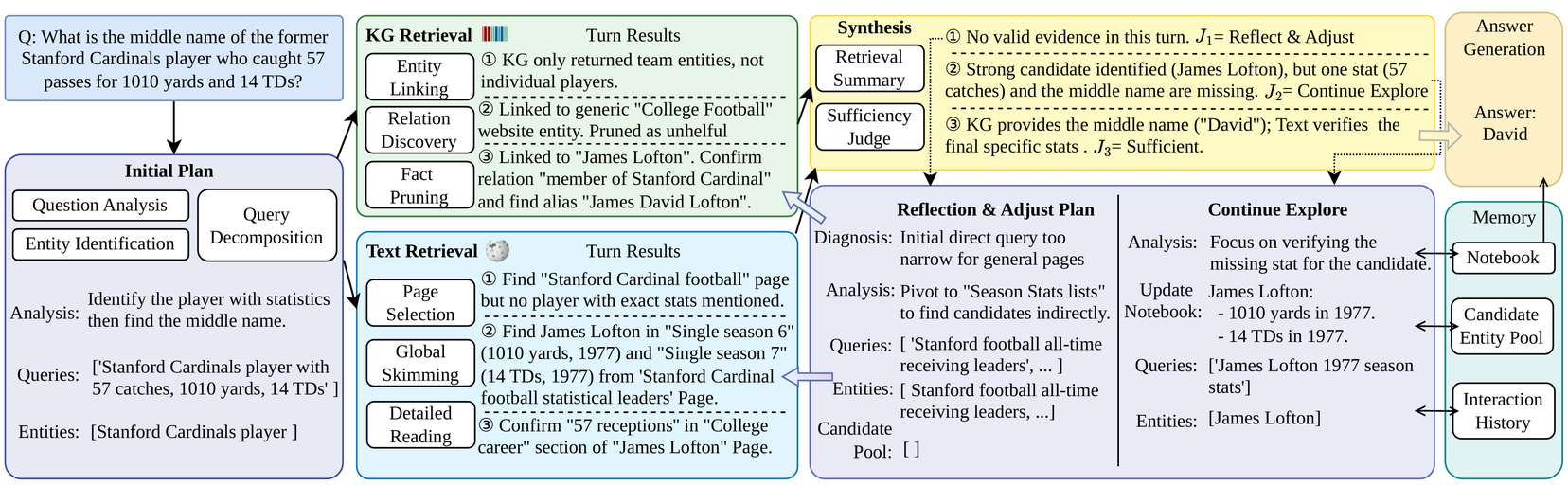
Key Contributions

Adaptive cognitive cycles. Training-free plan–explore–synthesize–reflect cycles for adaptive exploration.

Bidirectional graph–text synergy. Graph entities guide text reading, while text-discovered entities guide graph exploration.

Global-scale effectiveness. Validated across diverse benchmarks and backbone LLMs over full-scale Wikidata–Wikipedia.

CoG Framework



Adaptive Cognitive Cycle

CoG iteratively plans, explores KG and text, synthesizes evidence, and reflects on progress until sufficient evidence is gathered.

1	**Plan:**	Identify needs, sub-queries, and anchor entities.
2	**Explore:**	Retrieve KG facts and read entity documents.
3	**Synthesize:**	Combine graph–text evidence and assess sufficiency.
4	**Reflect:**	Refine the plan as evidence grows, or redirect exploration at dead ends.

Memory: A notebook, candidate entity pool, and interaction history retain evidence and guide future exploration.

Main Results

Baseline Type	Method	Multi-hop KG-based QA				Multi-hop text-based QA		
		KGQAGen	CWQ	QALD10-en	WebQSP	2WikiMQA	AdvHotpotQA	MusiQue
LLM-only	Direct	36.70	32.42	47.45	72.45	47.20	20.13	8.40
	CoT	38.28	33.59	45.35	70.84	47.40	23.70	9.60
	CoT-SC	40.50	33.40	46.55	71.40	49.60	24.03	10.60
Text-based RAG	Vector RAG	48.66	30.76	39.04	70.49	52.60	32.14	12.40
	ReAct	50.79	34.18	41.44	76.15	63.20	41.88	17.00
	IRCoT	53.75	33.30	41.14	71.40	71.60	47.08	20.60
	Search-o1	46.34	36.72	43.54	72.03	44.60	40.26	19.40
KG-based RAG	ToG	46.90	28.42	48.65	73.28	50.40	18.83	9.20
	PoG	50.93	27.15	48.95	78.53	73.20	15.26	14.00
	DoG	54.77	33.11	**55.86**	**82.38**	61.20	20.13	11.40
Hybrid RAG	ToG-2	55.61	41.41	53.45	81.89	73.00	35.71	16.20
Proposed	CoG	**74.70**	**42.87**	55.26	81.75	**85.20**	**51.95**	**27.80**

Table 2: Main results (Exact Match, %) of different methods on seven multi-hop QA datasets. All methods use Qwen3-32B (non-thinking mode) as the backbone LLM. The best results are highlighted in **bold**.

Baseline Type	Method	Multi-hop KG-based QA				Multi-hop text-based QA			Avg.
		KGQAGen	CWQ	QALD10-en	WebQSP	2WikiMQA	AdvHotpotQA	MusiQue	
LLM-only	Direct	38.74	34.57	40.84	64.06	29.20	23.38	14.20	35.00
	CoT	38.28	35.06	44.14	60.28	31.20	25.97	15.40	35.76
	CoT-SC	39.48	32.71	43.54	61.05	32.40	26.62	14.80	35.80
Text-based RAG	Vector RAG	50.23	33.20	37.84	64.83	48.80	45.45	18.20	42.65
	ReAct	48.19	35.74	42.04	69.79	57.60	54.87	22.80	47.29
	IRCoT	52.55	34.57	37.84	66.43	68.60	61.69	25.00	49.53
	Search-o1	52.27	39.84	48.65	68.60	47.80	47.73	27.40	47.47
KG-based RAG	ToG	42.63	29.20	45.95	62.87	34.80	22.08	11.20	35.53
	PoG	51.44	29.59	49.85	70.42	79.20	19.80	16.40	45.24
	DoG	43.65	25.68	51.65	63.29	62.00	24.68	13.80	40.68
Hybrid RAG	ToG-2	53.57	41.50	49.85	70.21	65.20	42.86	22.40	49.37
Proposed	CoG	**66.82**	**44.92**	**59.76**	**76.57**	**88.20**	**62.34**	**32.80**	**61.63**

Table 7: LLM-as-judge CORRECT rate (%) on seven datasets. The best result in each column is highlighted in **bold**.

Wikidata 89.3M entities 1.38B edges

Wikipedia ≈7.1M articles >5B words

Full Wikidata and complete entity-centric Wikipedia articles, without oracle paragraph selection.

Experimental Setup

Backbone: Qwen3-32B, non-thinking.
Evaluation criteria: Exact Match (%) and reference-based LLM-as-judge.

Strong across graph and text reasoning

Broad effectiveness across entity-centric KG questions and text-intensive multi-hop reasoning.

Strong gains on challenging tasks

Substantial EM gains over the strongest baselines: +19.09 pts on KGQAGen, +7.20 pts on MusiQue.

Consistent across evaluation metrics

Both evaluations support effectiveness: best results on 5/7 (EM) and 7/7 (LLM-as-judge) benchmarks.

Ablation & Exploration Depth

Variant	KGQA	CWQ	QALD	2Wiki	AdvHot	MusiQ	KG Avg.	Text Avg.	Avg.
CoG (Full)	74.70	**42.87**	**55.26**	**85.20**	**51.95**	**27.80**	57.61	**54.98**	**56.30**
Impact of Knowledge Sources									
w/o Text	**75.90**	39.06	51.65	76.20	27.27	20.20	55.54	41.22	48.38
w/o KG	56.44	40.92	48.65	79.80	49.68	20.80	48.67	50.09	49.38
Impact of Bidirectional Synergy									
w/o T-guide	74.14	42.58	52.85	76.20	39.61	22.80	56.52	46.20	51.36
w/o Synthesis	73.12	41.11	50.15	83.60	43.83	24.20	54.79	50.54	52.67
Impact of Cognitive Strategy									
w/o Reflect	72.57	42.38	52.85	85.00	48.05	25.80	55.93	52.95	54.44
w/o Plan	78.78	43.85	50.75	81.00	47.73	25.40	**57.79**	51.38	54.59
w/o Adap.EL	70.81	44.14	53.15	86.00	47.73	23.40	56.03	52.38	54.21

Table 3: Ablation results (EM, %) of CoG across three groups: (1) **Knowledge Sources**: removing either KG (w/o KG) or Text (w/o Text); (2) **Bidirectional Synergy**: disabling text-guided KG exploration (w/o T-guide), or removing summary synthesis module (w/o Synthesis); (3) **Cognitive Strategy**: excluding reflection-based adjustment (w/o Reflect), proactive planning (w/o Plan), or adaptive entity linking (w/o Adap.EL).

Complementary knowledge: KG and text both improve reasoning (+6.92 / +7.92 EM points).

Text-to-KG guidance: Text-discovered entities provide new graph anchors to bridge knowledge gaps (+4.94 EM points).

Adaptive control: Planning, reflection, and adaptive entity linking support effective exploration (+1.71 / +1.86 / +2.09 points).

Method	KG-based QA					Text-based QA				Avg.
	KGQAGen	CWQ	QALD10	WebQSP	Avg.	2Wiki	AdvHotpot	MusiQue	Avg.	
VectorRAG	48.66	30.76	39.04	70.49	47.24	52.60	32.14	12.40	32.38	40.87
CoG (Depth=1)	67.47	37.99	45.35	80.63	57.86	48.00	26.62	4.60	26.41	44.38
CoG (Depth=2)	71.83	42.58	51.95	81.05	61.85	85.20	46.43	20.00	50.54	57.01
CoG (Depth=3)	74.24	42.58	53.75	**81.75**	63.08	**86.80**	49.35	25.00	53.72	59.07
CoG (Depth=4)	74.70	42.87	**55.26**	**81.75**	**63.65**	85.20	51.95	**27.80**	**54.98**	**59.93**
CoG (Depth=5)	74.51	42.97	51.65	81.47	62.65	84.00	52.92	26.00	54.31	59.07
CoG (Depth=6)	**75.90**	**43.65**	54.35	80.56	63.62	83.20	**53.25**	25.60	54.02	59.50

Depth 4 gives the best overall EM. Optimal depth varies by task.

Generalization

Backbone	Method	KG Avg.	Text Avg.	Overall Avg.
Qwen3-32B	Direct	41.48	25.24	33.36
	IRCoT	42.73	46.43	44.58
	Search-o1	42.20	34.75	38.48
	DoG	47.93	30.90	39.42
	ToG-2	50.17	41.63	45.90
	CoG	**59.31**	**54.98**	**57.15** (+71.3%)
Qwen3-8B	Direct	31.09	16.73	23.91
	CoG	**53.39**	**48.55**	**50.97** (+113.2%)
gpt-oss-120b	Direct	28.33	15.88	22.10
	CoG	**42.87**	**40.28**	**41.58** (+88.1%)
DeepSeek-V3.2	Direct	47.91	28.06	37.99
	CoG	**64.79**	**60.50**	**62.64** (+64.9%)
GPT-5.1	Direct	45.67	38.67	42.17
	CoG	**56.00**	**55.33**	**55.67** (+32.0%)

Across backbones: Consistent gains over Direct Prompting (+32.0%–113.2% relative improvement).

Strong with smaller models: CoG-8B surpasses strongest 32B baselines (50.97% vs. 45.90% EM).

Takeaway

Adaptive cognitive cycles and text-guided graph exploration enable effective multi-hop QA over global knowledge bases.

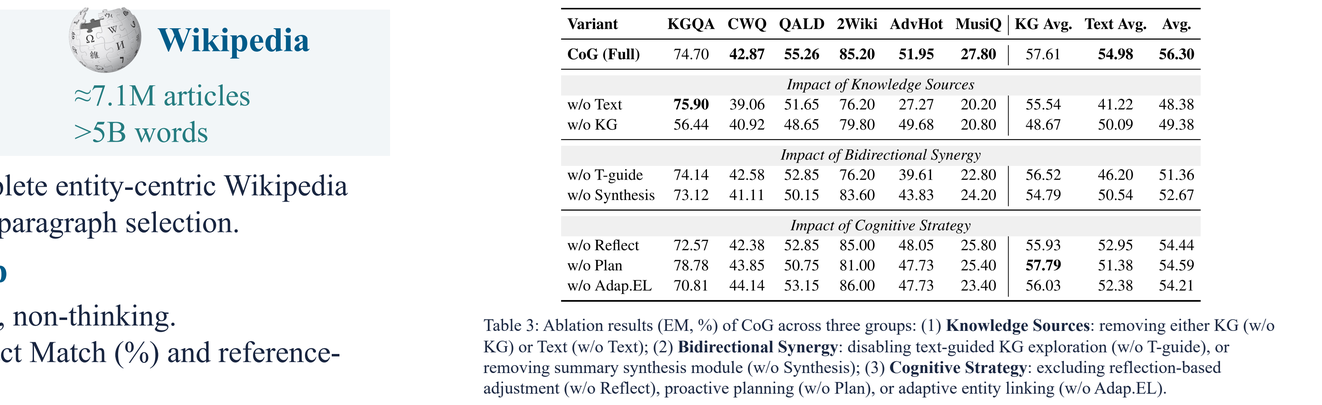
Case Study: Recovering the Stanford Evidence Chain

Question

What is the **middle name** of the former Stanford player who caught **57 passes** for **1,010 yards** and **14 TDs**?

Key insight

CoG redirects failed searches through reflection, turns text discoveries into graph anchors, and jointly verifies the answer using KG and text.



Contact

Gengxian Zhou

Research on knowledge-augmented LLMs for complex scientific problems, focusing on scientific knowledge representation, retrieval, and reasoning, with applications to extreme-weather understanding and attribution.

Research interests: Knowledge-Augmented LLMs · GraphRAG · Scientific Knowledge & Evaluation

Email: qq1679321454@live.cn

WeChat · Scholar · Paper · Code · Dataset